

# Community Detection on Signed Graphs: A Bayesian MCMC Coupling Approach



Seminar of the MathNet team (Inria Paris)

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# Outline

1. Motivation & the Sankararaman–Baccelli Model
2. The Bayesian Framework & Coupling
3. MCMC Dynamics & Performance Bounds
4. Open Questions & Conclusion

# 01

## Motivation & the Sankararaman–Bacelli Model



# Haplotype Assembly

## 1. Biological Setup: Two Homologous Chromosomes (Haplotypes)

Humans are diploid: they have two copies of each chromosome (homologous chromosomes).

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**Haplotype 1**  
(Truth  $\Sigma = +$ )

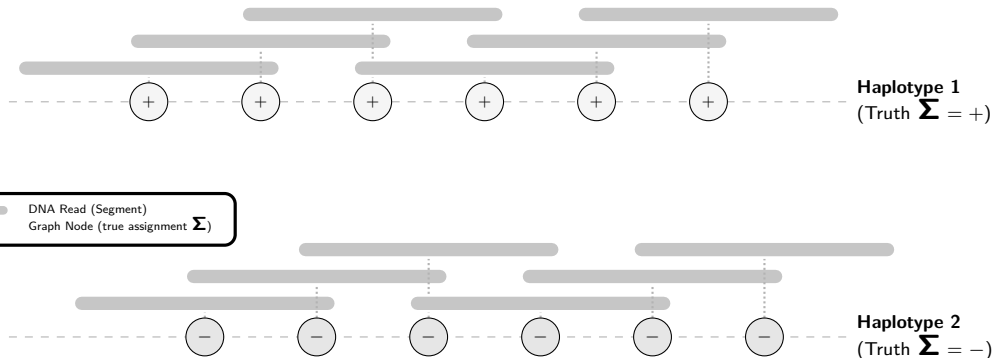
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**Haplotype 2**  
(Truth  $\Sigma = -$ )



# Haplotype Assembly

## 2. High-Throughput Sequencing: DNA Reads

Sequencing generates short, overlapping fragments called **reads** from both chromosomes.



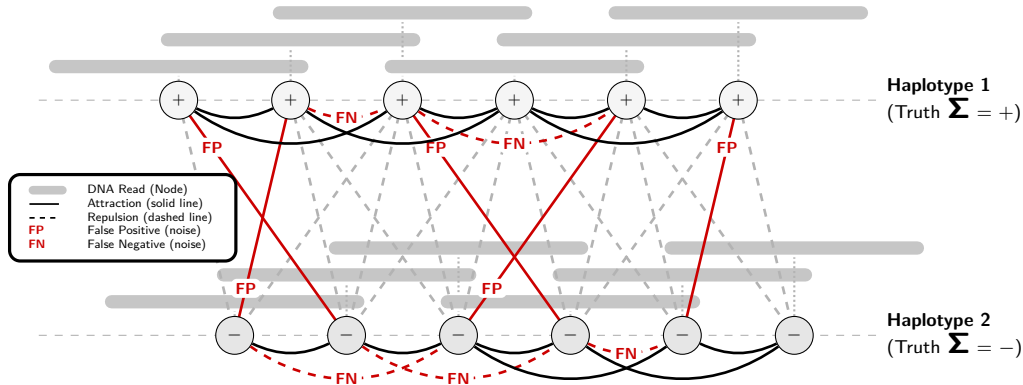


# Haplotype Assembly

## 3. Modeling as a Signed Graph

Nodes = reads      Edges = overlaps

**Goal:** Reconstruct the true haplotype assignments  $\Sigma$  from this observed signed graph.





## Motivation: Signed Community Detection

### Community Detection on Signed Graphs ( $K = 2$ )

- ▶ We aim to partition nodes into  $K = 2$  communities:  $\Sigma_i \in \{+1, -1\}$ .
- ▶ Observed links represent **attractions** (solid lines) and **repulsions** (dashed lines).
- ▶ Under the hood, this is a **Stochastic Block Model** (SBM) setup.

### The Geometric Challenge

- ▶ Nodes are embedded in space (e.g., genomic positions on  $\mathbb{Z}$  or  $\mathbb{R}^d$ ).
- ▶ Edge probabilities depend directly on the distance  $r = \|x_i - x_j\|$ :

$$\mathbb{P}(\text{attractive edge} \mid \text{same community}) = f_{\text{in}}(r)$$

$$\mathbb{P}(\text{repulsive edge} \mid \text{different communities}) = f_{\text{out}}(r)$$

- ▶ **All the difficulty (and the interest!) comes from this geometric dependency.**



# The Sankararaman–Baccelli Model (GSBM)

## Model Definition [1]

- ▶ **Spatial Nodes**  $V_n$ : Hom. POISSON PP  $\mathcal{X}$  in  $\mathbb{R}^d$  [2] restrict. to a  $n$ -volume window.
- ▶ **Latent Communities**: Uniform assignments  $\Sigma_i \in \{+1, -1\}$  ( $K = 2$ ).
- ▶ **Geometric Overlaps**: Observed connection  $e = \{i, j\} \in H$  with probability:

$$\mathbb{P}(e \in H \mid X, \Sigma) = \begin{cases} f_{\text{in}}(r_e) & \text{if } \Sigma_i = \Sigma_j \text{ (attraction),} \\ f_{\text{out}}(r_e) & \text{if } \Sigma_i \neq \Sigma_j \text{ (repulsion),} \end{cases}$$

with distance  $r_e = \|X_i - X_j\|$  and connecting functions  $f_{\text{in}}(r) \geq f_{\text{out}}(r)$ .

## Goal: Weak Recovery

- ▶ Find an algorithm  $\tau$  (independent of  $\Sigma$ ) reconstructing assignments:

$$\lim_{n \rightarrow \infty} \mathbb{E}[\text{ov}_n(\Sigma, \tau)] \geq \frac{1}{2} + \eta \quad \text{for some } \eta > 0,$$

using the **Community Overlap**:  $\text{ov}_n(\Sigma, \tau) = \max_{\pi \in \mathfrak{S}_2} \frac{1}{n} \sum_{i=1}^n \mathbb{I}_{\{\tau_i = \pi(\Sigma_i)\}}.$



# Gibbs Model on Signed Graphs

## Attractive and Repulsive Edges

We partition the potential edges  $E = F \dot{\cup} \tilde{F}$ :

- ▶  $F$ : **Attractive** (ferromagnetic) edges, with weight  $W_e > 0$
- ▶  $\tilde{F}$ : **Repulsive** (antiferromagnetic) edges, with weight  $W_e < 0$

## Signed Gibbs Measure [3]

The probability of a label configuration  $\sigma \in \{+1, -1\}^{V_n}$  is  $\mu(\sigma) = \frac{1}{Z} e^{-U(\sigma)}$ , where the energy  $U(\sigma)$  penalizes unsatisfied edges:

$$U(\sigma) = \sum_{e=\{i,j\} \in F} W_e \mathbb{I}_{\{\sigma_i \neq \sigma_j\}} + \sum_{e=\{i,j\} \in \tilde{F}} |W_e| \mathbb{I}_{\{\sigma_i = \sigma_j\}}$$

The magnitudes  $|W_e|$  determine the strength of each constraint.



# From GSBM to the Gibbs Posterior

## Conditioning on the Point Cloud

- ▶ Once the spatial positions  $X$  are observed and fixed, all pairwise distances  $r_e = \|X_i - X_j\|$  are known.
- ▶ Under a uniform prior on  $\Sigma$ , the posterior distribution given the observed graph  $H$  is:

$$\mathbb{P}(\Sigma = \sigma \mid H, X) \propto \mathbb{P}(H \mid X, \sigma)$$

## Log-Likelihood as Signed Gibbs Energy

- ▶ Factorizing the likelihood over all potential edges  $e = \{i, j\} \in E$ :

$$\mathbb{P}(H \mid X, \sigma) \propto e^{-U(\sigma)}$$

- ▶ The posterior  $\mathbb{P}(\Sigma \mid H, X)$  is exactly a **Signed Gibbs measure** with weights:

$$W_e(H) = \begin{cases} \log\left(\frac{f_{\text{in}}(r_e)}{f_{\text{out}}(r_e)}\right) > 0 & \text{if } e \in H \text{ (observed overlap / attraction } \in F), \\ \log\left(\frac{1 - f_{\text{out}}(r_e)}{1 - f_{\text{in}}(r_e)}\right) < 0 & \text{if } e \notin H \text{ (absent overlap / repulsion } \in \tilde{F}). \end{cases}$$



# The Percolation Obstruction

## Necessary Condition for Weak Recovery [1]

- ▶ Intuitively, if the graph is composed of many tiny, isolated components, no global community structure can be reconstructed.
- ▶ Information can only flow globally if the graph **percolates** (possesses a giant/infinite component).

## Theorem (Sankararaman & Baccelli [1])

*Percolation of the graph with edge probability*

$$p_e = f_{\text{in}}(r_e) - f_{\text{out}}(r_e)$$

*is a **necessary condition** for weak recovery. If this independent bond percolation does not yield a giant component, weak recovery is mathematically impossible.*

- ▶ Our goal: Retrieve and extend this result using a unified **Bayesian MCMC coupling** framework.

02

# The Bayesian Framework & Coupling





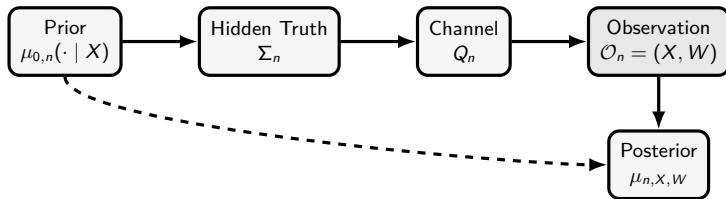
# The General Bayesian Model

## Definition

General Bayesian Community Detection Model

1. **Features:**  $X_n \in \mathcal{X}_n$  with distribution  $\nu_n$ .
2. **Prior:**  $\mu_{0,n}(\mathrm{d}\sigma \mid x)$  over  $\mathcal{C}_n = \mathcal{L}_K^{V_n}$ .
3. **Channel:**  $Q_n(\mathrm{d}w \mid x, \sigma) = \prod_{e \in E_n} Q_{e,n}(\mathrm{d}w_e \mid x_i, x_j, \sigma_i, \sigma_j)$ .

The joint law is  $\mathbb{P}_n(\mathrm{d}x, \mathrm{d}\sigma, \mathrm{d}w) = \nu_n(\mathrm{d}x) \mu_{0,n}(\mathrm{d}\sigma \mid x) Q_n(\mathrm{d}w \mid x, \sigma)$ .





## Posterior Resampling (Nishimori Identity)

### Replacing Truth by a Replica

- ▶ Since  $\Sigma_n$  is hidden, we cannot compute the overlap  $\text{ov}_n(\Sigma_n, \tau_n)$  directly in an algorithm.
- ▶ However, under the true model, the hidden truth and a sample from the posterior are exchangeable.

### Theorem (Posterior Resampling / Nishimori Identity [4])

Let  $\tau_n$  be any community detection algorithm. Let  $\Sigma_n^{(1)} \sim \mu_{n, X_n, W_n}$  be a replica drawn independently from the posterior. For any bounded test function  $\Psi_n$ :

$$\mathbb{E} [\Psi_n(X_n, W_n, \Sigma_n, \tau_n)] = \mathbb{E} [\Psi_n(X_n, W_n, \Sigma_n^{(1)}, \tau_n)] .$$

### Corollary

*Overlap Identity* For any algorithm  $\tau_n$  and threshold  $s \in [0, 1]$ :

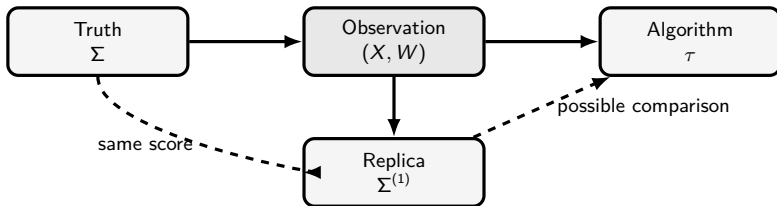
$$\mathbb{P} [\text{ov}_n(\Sigma_n, \tau_n) \geq s] = \mathbb{P} [\text{ov}_n(\Sigma_n^{(1)}, \tau_n) \geq s] .$$



# Invariant Coupling

## Markov Chain Dynamics as a Source of Coupling

- ▶ To study the overlap, we need a way to construct the posterior replica  $\Sigma^{(1)}$  starting from the ground truth  $\Sigma$ .
- ▶ Let  $M_{n,x,w}$  be a MARKOV transition kernel that leaves the posterior  $\mu_{n,x,w}$  invariant.
- ▶ If we initialize the chain at the hidden truth  $\Sigma$ , and perform **exactly one step** of the dynamics to obtain  $\Sigma^{(1)} \sim M_{n,x,w}(\Sigma, \cdot)$ , then  $\Sigma^{(1)}$  is still distributed according to the posterior!



# 03

## MCMC Dynamics & Performance Bounds



# MCMC Dynamics for Signed Gibbs Measures

## 1. Single-Spin Dynamics (Glauber / Metropolis–Hastings)

- ▶ Select a node at random and update its label according to its local neighbors.
- ▶ Extremely slow in the presence of frustration: local updates cannot cross energy barriers.

## 2. Swendsen–Wang (SW) Signed Cluster Dynamics [5, 6]

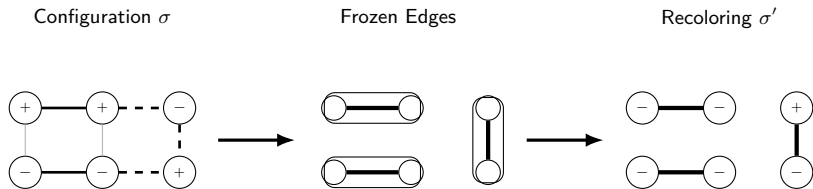
- ▶ Update whole clusters of vertices simultaneously to speed up mixing.
- ▶ **Freezing Step:** For each edge  $e = \{i, j\}$ :
  - If  $e$  is satisfied ( $W_e > 0$  and  $\sigma_i = \sigma_j$ , or  $W_e < 0$  and  $\sigma_i \neq \sigma_j$ ), freeze it with probability:

$$p_e^{\text{gel}} = 1 - e^{-|W_e|}.$$

- If  $e$  is unsatisfied, do not freeze it ( $p_e^{\text{gel}} = 0$ ).
- ▶ **Recoloring Step:** Flip each frozen cluster independently with probability  $1/2$ .



## One Step of Signed Swendsen–Wang



- ▶ Edges in bold represent satisfied interactions that are randomly frozen.
- ▶ Clusters are then flipped independently.



## Coupling & the Sankararaman–Bacelli Bound

### Gel Probability under the Generative Model

Let's run one step of SW starting from the hidden truth  $\Sigma$ . What is the probability that an edge  $e$  is frozen?

- **Case 1:**  $\Sigma_i = \Sigma_j$ . Edge is present with probability  $f_{\text{in}}(r_e)$ , and has weight  $W_e = \log(f_{\text{in}}/f_{\text{out}}) > 0$ .

$$\mathbb{P}(\text{frozen}) = f_{\text{in}}(r_e) \left(1 - e^{-|W_e|}\right) = f_{\text{in}}(r_e) \left(1 - \frac{f_{\text{out}}(r_e)}{f_{\text{in}}(r_e)}\right) = f_{\text{in}}(r_e) - f_{\text{out}}(r_e).$$

- **Case 2:**  $\Sigma_i \neq \Sigma_j$ . Edge is absent with probability  $1 - f_{\text{out}}(r_e)$ , and has weight  $W_e = -\log((1 - f_{\text{out}})/(1 - f_{\text{in}})) < 0$ .

$$\mathbb{P}(\text{frozen}) = (1 - f_{\text{out}}(r_e)) \left(1 - \frac{1 - f_{\text{in}}(r_e)}{1 - f_{\text{out}}(r_e)}\right) = f_{\text{in}}(r_e) - f_{\text{out}}(r_e).$$

### Corollary

*Independent Bond Percolation Under the true generative model, the graph of frozen edges is **exactly** an independent bond percolation with edge probability:*



# High-Order Triangular Dynamics

## The Frustration Obstruction

- ▶ Standard edge-by-edge SW freezes satisfied edges independently.
- ▶ In frustrated systems, this leads to conflicts and blocks dynamics due to excessive freezing.
- ▶ **Idea:** Introduce correlations on the freezing mechanism using higher-order cliques (**triangles**) to freeze as little as possible.

## 3. Triangular Cluster Dynamics

- ▶ Transfer edge energy to triangles:  $U(\sigma) = \sum_{t \in \mathcal{T}} U_t(\sigma)$ .
- ▶ **Attractive Triangle** (+ + + or - - -): Freeze all 3 edges (prob  $a_u = 1 - e^{-2u}$ ) or none (prob  $b_u = e^{-2u}$ ).
- ▶ **Repulsive Triangle** (2 satisfied edges, 1 unsatisfied): Freeze exactly one of the two satisfied edges with probability  $a_u/2$ . This keeps the system in a low energy state with minimal freezing.



## Triangular Dynamics Freezing Rules

| $\triangle$ <b>Attractive Triangle (Isotropic)</b> |             |          |          |          |                  |
|----------------------------------------------------|-------------|----------|----------|----------|------------------|
| Configuration                                      | $\emptyset$ | $\{ab\}$ | $\{ac\}$ | $\{bc\}$ | $\{ab, ac, bc\}$ |
| $(+, +, +)$                                        | $b_u$       | 0        | 0        | 0        | $a_u$            |
| Other                                              | 1           | 0        | 0        | 0        | 0                |

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| $\triangleleft \triangle$ <b>Repulsive Triangle (Isotropic)</b> |             |          |          |          |                  |
|-----------------------------------------------------------------|-------------|----------|----------|----------|------------------|
| Configuration                                                   | $\emptyset$ | $\{ab\}$ | $\{ac\}$ | $\{bc\}$ | $\{ab, ac, bc\}$ |
| $(+, +, +)$                                                     | 1           | 0        | 0        | 0        | 0                |
| $(+, +, -)$                                                     | $b_u$       | 0        | $a_u/2$  | $a_u/2$  | 0                |

**Table:** Freezing rules for triangular dynamics ( $a_u = 1 - e^{-2u}$ ,  $b_u = e^{-2u}$ ). By construction, these rules satisfy local reversibility and leave the posterior distribution invariant.



## Strict Improvement on the Triangular Grid

### Homogeneous Model ( $f_{\text{in}} = p$ , $f_{\text{out}} = 1 - p$ )

On the triangular grid, we select a family of independent white triangles (like a chessboard).

- ▶ **Edge Dynamics (Standard SW):** Gelled graph is independent edge percolation with parameter  $p_{\text{edge}} = 2p - 1$ . Critical threshold:  $p_c^{\text{edge}} = \frac{1}{2} + \sin(\frac{\pi}{18}) \approx 0.673$ .
- ▶ **Triangular Dynamics:** Gelled graph yields correlated triangle percolation. Critical threshold:  $p_c^{\Delta} = \frac{7}{4} - \frac{\sqrt{17}}{4} \approx 0.7192$ .

### Corollary

*Strictly Better Impossibility Bound [7] Since  $p_c^{\text{edge}} < p_c^{\Delta}$ , for any*

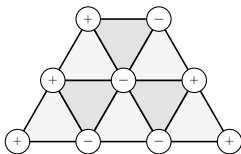
$$p \in [0.673; 0.7192[ ,$$

*standard edge dynamics percole (no conclusion), whereas triangular dynamics do not. Hence, we prove that weak recovery is impossible in this wider range of parameters!*



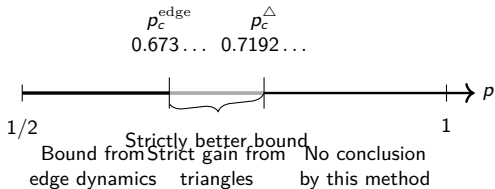
# Comparison of Percolation Thresholds

Triangular Grid



White triangles used by  
triangular dynamics

Weak Recovery Impossibility Bounds



# 04

## Open Questions & Conclusion





## Open Questions

### 1. Extension to Other Lattices and Higher Dimensions

- ▶ What are the thresholds on the Kagome, FCC, HCP, or Hyper-cubic lattices?
- ▶ Can we define tetrahedral or higher-order simplex dynamics to obtain even tighter bounds?

### 2. Multi-Community case ( $K > 2$ )

- ▶ How does the cluster dynamics extend when the spins take  $K$  values?
- ▶ What is the role of Potts-like percolation in the coupling?

### 3. Sufficiency of the Percolation Condition

- ▶ The percolation threshold provides a necessary condition (impossibility).
- ▶ Is percolation also sufficient for the existence of a reconstruction algorithm?

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*Merci.*

